Day-4: Understanding the Startup Ecosystem

Have you ever wondered why some cities are buzzing with startups and caught up in “hustle culture” while others lag behind?

That’s because startups rarely grow in isolation. They grow inside an ecosystem, a living network of people, institutions and resources that help new ideas turn into real businesses. So what even is a startup ecosystem?

A **startup ecosystem** is a community of entrepreneurs, investors, mentors, support organisations and government policies that work together to help new ideas grow. Think of it like a rainforest: a small sapling (the startup) needs sunlight (funding), nutrients (mentors and talent) and protection (laws and support systems) to grow into a tall tree.

Now let’s look at the exact pieces that make up this ecosystem. Alongside each one, you’ll see how it shows up in the story of **Razorpay**, an Indian payments startup we will keep coming back to all through this module.

Key Elements of a Startup Ecosystem

Element	Description	Example
Startups	The innovators and risk-takers building new products or services.	Razorpay, Zomato
Investors	Provide the money: angel investors, VCs and big funds.	Sequoia (now Peak XV), Tiger Global, GIC
Incubators and Accelerators	Offer mentorship, network and seed funding so startups grow faster.	Y Combinator (backed Razorpay early on), T-Hub Hyderabad
Corporates	Partner with, buy from or acquire startups to stay innovative.	Zomato, Swiggy and Ola became early Razorpay customers
Universities and Research Institutes	Source of fresh ideas, research and technical talent.	IITs, IISc. Razorpay’s founders are IIT Roorkee alumni
Government and Regulators	Make the rules, give grants and promote entrepreneurship.	Startup India, DPIIT, RBI (regulates payment companies)
Support Services	Legal, HR, design and accounting firms that specialise in startups.	LegalDesk, MyStartupCFO

Key Players in the Ecosystem

- **Entrepreneurs:** visionaries who spot a real problem and decide to solve it.
- **Investors:** individuals or firms who back promising startups with money and support.
- **Mentors and Advisors:** people who have played the game before and help founders avoid common mistakes.

- **Accelerators and Incubators:** programmes that nurture early-stage startups through the tough first months.
- **Government Agencies:** provide policy support, funding schemes and startup-friendly laws.
- **Coworking Spaces:** affordable spaces where founders meet, collaborate and network.

A strong ecosystem lowers the barriers to starting up, encourages people to share knowledge, and raises everyone's chances of success. That is why cities like Bengaluru, San Francisco and Singapore attract founders: they don't just offer funding, they offer a community.

Market Analysis: Seeing the Bigger Picture

Imagine you've built a great product. Before you pitch it to an investor, they will ask one big question:

"How big is your market, and can you make money from it?"

That is where market analysis comes in. It helps you understand who your customers are, how many of them exist, and whether your business can grow in a healthy way. We will cover three things: market size (TAM, SAM, SOM), unit economics, and competitive moats.

1. TAM, SAM and SOM: Measuring Market Size

Term	What it means	Simple example (smart water bottle)
TAM (Total Addressable Market)	The entire demand for your kind of product or service.	Everyone in the world who drinks water
SAM (Serviceable Available Market)	The part of TAM that your business model can actually serve.	People in India who buy fitness gadgets
SOM (Serviceable Obtainable Market)	The share of SAM you can realistically win in the near future.	The top 5% most health-conscious urban buyers

Think of it like this: if TAM is the universe, SAM is your galaxy and SOM is the planet you are actually going to land on. Investors love clarity here. It tells them whether you are fishing in a small pond or an ocean.

Worked example: market sizing for Razorpay

Razorpay was started in 2014 by Harshil Mathur and Shashank Kumar. Its main product is a payment gateway, which lets a business accept online payments (cards, UPI, netbanking) and pays Razorpay a small fee on each payment. So the market we are measuring is **the money Indian businesses pay for payment services in a year.** We will do it bottom-up, which means counting customers and multiplying by what each one spends.

Note: the numbers below are rough assumptions made for teaching. A real analyst would check each one against reports from RBI, NPCI or the government before using it.

Step 1, TAM. Assume India has about 6 crore small and medium businesses, and each could pay around ₹15,000 a year in payment fees.

6 crore × ₹15,000 = **₹90,000 crore**

Step 2, SAM. Not every shop is ready for online payments. Say only 1 in 4 is registered and digitally active, which is 1.5 crore businesses.

1.5 crore × ₹15,000 = **₹22,500 crore**

Step 3, SOM. Razorpay has a strong brand, lots of developers using it and a big sales team, so say it can realistically win about 25% of SAM in the next 2 to 3 years.

25% of ₹22,500 crore = about **₹5,600 crore**

Sanity check. Razorpay's actual revenue for FY25 was about ₹3,783 crore. That is roughly 17% of our SAM, so a SOM of 25% looks believable. If your SOM comes out far lower than what the company already earns, or bigger than the whole SAM, go back and fix your assumptions.

Your turn: market sizing exercise

Pick any real startup you like (for example Zomato, Lenskart, Cult.fit or Physics Wallah) and size its market in 4 steps:

- Write down who the customer is and what they pay for.
- Estimate how many such customers exist (TAM) and how much each one spends in a year.
- Narrow it down to the customers the startup can really serve (SAM), for example by city, price or product.
- Pick a realistic share you can win (SOM), then compare it with the startup's real revenue or user numbers as a sanity check.

Layer	Who is included	Calculation	Size
TAM			
SAM			
SOM			

2. Unit Economics: The Math Behind Each Sale

Now that you know how big the market is, the next question is: **"Does your business make a profit on each customer?"** Unit economics means looking at the money earned and spent per unit, where a unit can be a product, a subscription or a customer.

Two key metrics:

- **CAC (Customer Acquisition Cost):** how much you spend to get one new customer.
- **LTV (Customer Lifetime Value):** how much profit that customer brings you over the whole time they stay.

Golden Rule: if LTV is more than 3 × CAC, your startup has healthy unit economics.

A quick example: if you spend ₹500 to get a customer and that customer brings you ₹2,000 in profit over their lifetime, LTV is 4 times CAC, so you are on the right track.

Razorpay example (assumed numbers). Take one mid-sized online store that uses Razorpay:

Step	Number
The store processes about ₹20 lakh of payments every month	₹20 lakh
Razorpay earns about 1.5% of that as a fee, so in a year	₹20 lakh × 1.5% × 12 = ₹3.6 lakh
Razorpay keeps about 34% after direct costs (FY25: gross profit ₹1,277 cr on revenue ₹3,783 cr)	₹3.6 lakh × 34% = about ₹1.22 lakh a year
If the store stays for 4 years, LTV is	₹1.22 lakh × 4 = about ₹4.9 lakh
Assume it costs ₹80,000 to find and onboard this store (CAC)	₹80,000
LTV ÷ CAC	₹4.9 lakh ÷ ₹80,000 = about 6, which is above 3 and healthy

Razorpay does not publish its real CAC or LTV, so these are only for practice. An investor would ask the founders for the real numbers.

3. Competitive Moats: Protecting What You Build

Once your business starts doing well, competitors will follow. That is why investors look for a **moat**, something that makes your startup hard to copy or replace. Just like a moat around a castle, it keeps the competition away. Moats turn a short-term lead into long-term strength. They are the difference between being a trend and being a titan.

Type of moat	Description	Example
Technology moat	Technology or patents that others cannot easily copy.	Tesla's battery tech; Razorpay's smart routing that raises payment success rates
Network effect	The product becomes more valuable as more people use it.	WhatsApp, LinkedIn
Brand moat	Customers trust your name.	Apple
Cost advantage	You can sell cheaper because you run efficiently at scale.	Amazon's logistics
Data advantage	Unique data keeps improving your product.	Google Search; Razorpay uses payment history to offer loans to merchants
Switching cost	It is painful for customers to leave.	A business using Razorpay for payments, payouts and banking together

Razorpay competes with PayU, BillDesk and Paytm, and these all offer similar services. A low price alone is easy to match, which is why moats like data and switching costs matter.

Startup Business Models

A business model simply answers three questions: **who pays, for what, and how often?** When investors look at a startup, this is often the first thing they check, because it decides how revenue grows.

Model	How it makes money	Example
Transaction fee	Takes a small cut of every payment or transaction that passes through it.	Razorpay (roughly 2% on card payments, lower on UPI)
Marketplace / commission	Connects buyers and sellers and charges a commission.	Zomato, Swiggy
SaaS (subscription)	Customers pay a monthly or yearly fee to use software.	Freshworks, Zoho
Freemium	The basic product is free, and advanced features cost money.	Duolingo, Canva
Direct to consumer (D2C)	The startup makes its own products and sells them online directly to customers.	Mamaearth, boAt

Notice how Razorpay earns more when its customers sell more. Its revenue grows with the volume of money flowing through it, so it does not need to charge a high price per payment. Different models suit different problems, and many startups mix two or more as they grow.

Quick Recap

- A startup ecosystem is a network of founders, investors, mentors, universities, government and support services.
- TAM, SAM and SOM tell you how big the opportunity really is, from the universe down to your planet.
- Healthy unit economics means LTV is more than 3 times CAC.
- A moat is what protects a startup from copycats.
- A business model explains who pays, for what, and how often.

Sources: Razorpay FY25 results as reported by Entrackr and Inc42; Razorpay funding and product details from FinTech Global and Moneycontrol. Market size and unit economics numbers are assumptions for teaching.

Day-5: Due Diligence and Building a Deal Pipeline

Yesterday we learned how big a market can be and how a startup makes money. Today we flip to the investor's chair. Imagine a founder just pitched you, and you loved it. Do you write the cheque right away?

Not a chance. Around 9 out of 10 startups fail, so before putting money in, a good investor checks whether what the founder said is actually true. That check is called **due diligence**, and finding the startups worth checking in the first place is called **building a pipeline**. These two skills are what today is about.

Due Diligence: Checking Before You Invest

Due diligence is a detailed investigation an investor does to confirm a startup's claims and spot risks before investing. Think of buying a second-hand bike. The seller says it runs perfectly, but you still check the papers, look at the engine and take it for a ride. Due diligence is the same thing, only the bike is a company and the price is crores.

Where does it sit in the funding process? Every deal moves through these steps:

Step	What happens
1. Deal sourcing	The VC finds startups through LinkedIn, referrals, demo days and databases.
2. Initial screening	A quick look at the pitch and market to decide if it is worth more time.
3. Due diligence	The deep check on team, numbers, product, legal papers and customers. This is today's topic.
4. Term sheet	Both sides agree on how much money, at what valuation, for how much ownership.
5. Investment committee	The partners of the fund vote on whether to go ahead.
6. Funding and support	Money is sent and the VC helps with mentoring and introductions.

Early on, due diligence can be a quick gut-check of the team and the market. Closer to a deal it becomes a multi-week project, and the findings are written up in a short report for the investment committee.

Types of Due Diligence

There are seven main areas an investor looks at. Here is each one, with what gets checked and how Razorpay would fare.

Type	What the investor checks	Razorpay example
Founder and team	Past experience, track record, skills that complement each other, references, commitment to the startup.	Same two IIT Roorkee co-founders, Harshil Mathur and Shashank Kumar, have led it since 2014.

Market	TAM and SAM, growth rate, customer segments, trends and rules that can change the market.	India's digital payments keep growing, but RBI rules for payment companies can shift.
Product and technology	Is the product really built and used? Is the tech secure and able to scale? Who owns the IP?	Handles heavy traffic from big clients. Check uptime and payment success rate.
Financial	Revenue, margins, cash, burn rate, whether the numbers in the pitch match the books.	FY25 revenue ₹3,783 cr, gross profit ₹1,277 cr, loss of ₹1,209 cr.
Legal and compliance	Company papers, who owns how much (cap table), contracts, licences, court cases.	Needs RBI approvals as a payment aggregator and a clean ownership structure.
Customer and traction	Are real people using and paying? Do they stay? Does one customer give most of the revenue?	Customers include Zomato, Swiggy, Ola and Cred, so check how much revenue depends on them.
Competition and moat	Who the rivals are, what makes the startup hard to copy, how easy it is for new players to enter.	Rivals are PayU, BillDesk and Paytm. Moats: data, many connected products, brand.

Key Metrics and KPIs to Verify

A KPI (Key Performance Indicator) is just a number that shows how a business is doing. The golden habit is to never trust a number until you have checked it in at least two places. Here are the ones investors look at most:

KPI	**How it is worked out**	**Good sign**	**Razorpay (public data)**
Revenue growth	(This year − last year) ÷ last year	Fast and steady	₹2,296 cr to ₹3,783 cr, up 65%
Gross margin	Gross profit ÷ revenue	Going up over time	₹1,277 cr ÷ ₹3,783 cr = about 34%
Burn rate and runway	Monthly cash spent; cash ÷ monthly burn = months left	Runway of 18 months or more	Company says its cash position is strong
LTV : CAC	Lifetime profit per customer ÷ cost to get them	3 or higher	Not public, so ask the founders
Churn and retention	Customers who leave vs. who stay	Low churn, high retention	Not public, so ask for cohort data
Total payment volume (TPV)	Total value of payments handled	Growing every quarter	About $180 billion a year (company-stated)

Notice some cells say "not public". That is normal. A big part of due diligence is asking for exactly those numbers and seeing whether the founders share them openly.

Practical Template: Due Diligence Checklist

Here is a simple checklist you can reuse for any startup. Mark each row with: **Yes** (checked and fine), **Follow up** (needs a closer look) or **Not found**. The Razorpay column is filled in as an example, using only public information.

Area	What to verify	Razorpay finding	Status
Team	Founders' background, references, equity and vesting	Founders unchanged since 2014; investor list is public	Yes
Market	TAM and SAM, growth, rules	Large market; payment rules can change	Follow up
Product and tech	Stage, uptime, security, IP	Live product at scale; uptime data needed	Follow up
Financial	Statements, burn, margins, unit economics	Revenue and gross profit match across news reports; loss mostly one-off	Yes
Legal	Cap table, contracts, licences, cases	Moved the parent company back to India in 2025; check filings	Follow up
Customers	Revenue by customer, churn, interviews	Big names as clients; no churn data public	Not found
Competition	Rivals, moat, barriers to entry	Crowded market; moat is breadth of products	Follow up

Note: the Razorpay figures here are from FY25 reports. Check for newer results before you reuse them.

Your turn: due diligence exercise

Pick a startup, fill the checklist below and check at least five KPIs against two sources each.

Area	What to verify	Your finding and source	Status
Team			
Market			
Product and tech			
Financial			
Legal			
Customers			
Competition			

Sourcing Data and Tools for Sourcing

You can't check a startup you haven't found. Sourcing means finding startups and collecting reliable information about them. The good news is that most of it is a few clicks away.

Tool	What it is good for	Cost
LinkedIn	Founder and team backgrounds, team size, hiring trends, and warm introductions through mutual contacts.	Free to use; premium for more search
Crunchbase	Funding rounds, investors, similar companies and lists by sector.	Basic is free; full data is paid
PitchBook	Detailed deal and valuation data used by professional funds.	Paid; some college libraries offer it
Tracxn	Sector maps and competitor lists, strong for Indian startups.	Limited free; full is paid
Entrackr and Inc42	News on funding and yearly financial results of Indian startups.	Free
MCA portal (RoC)	Official company filings and director details.	Search is free; documents cost a small fee

A tip for checking: every database can be out of date or wrong, because many are filled by the companies themselves. When two sources disagree, trust the official one (company filings) first, then the news, then databases.

Pipeline Building and Lead Tracking

A **deal pipeline** is a running list of startups that are at different stages, from “just found” to “invested” or “passed”. Without one, good startups get forgotten and promises to follow up are missed. A lead tracker can be as simple as a spreadsheet with these columns: startup name, sector, source (where you found it), stage, score, last contact and next action.

To screen quickly, give every startup a score out of 10 on what VCs care about most:

What	Weight	Quick question to ask
Team	25%	Can these founders handle problems and change direction when needed?
Market size	20%	Is the SAM big enough for a really big outcome?
Product edge	20%	What makes it hard to copy?
Traction	20%	Are there real users, growth or revenue?
Unit economics	15%	Does each customer make money, with LTV above 3 times CAC?

Multiply each score by its weight and add them up. That gives you one score per startup so you can rank them.

Worked example: a mock pipeline

Say you are screening Indian payments startups. You search Crunchbase and LinkedIn, then log what you find. **The stages and scores below are made up for practice. They are not real opinions about these companies.**

Startup	Sector	Source	Stage	Score	Next action
Razorpay	Payment gateway	Crunchbase	Due diligence	8.5	Ask for cohort data
Cashfree Payments	Payment gateway	Tracxn	Screening	7.0	Pull financials
Juspay	Payments tech	LinkedIn	Screening	7.5	Call a founder
Setu	Fintech APIs	Entrackr	Sourced	6.5	Size the market
Decentro	Banking APIs	LinkedIn	Sourced	6.0	Check customers

Check how Razorpay's score might be built: team 9, market 8, product edge 8, traction 9, unit economics 8. That gives 9×0.25 + 8×0.20 + 8×0.20 + 9×0.20 + 8×0.15 = **8.5**.

Your turn: build your own pipeline

Choose one sector (EdTech, Climate Tech, HealthTech, anything you like). Use Crunchbase and LinkedIn to find at least 8 startups and fill in the tracker:

Startup	Sector	Source	Stage	Score	Next action

Power of Networking

Here is a secret about venture capital: many of the best deals never show up on any database. They come through a trusted introduction. Networking is how you hear about startups early, check what founders claim by speaking to people who know them, and get your own foot in the door.

- **Give before you ask.** Share a useful article, an introduction or honest feedback before asking for anything.
- **Go where startups are.** Demo days, college E-Cells, incubator events and LinkedIn communities.
- **Follow up in 2 days.** A short thank-you with one specific thing you learned goes a long way.

A message you could send on LinkedIn:

"Hi Priya, I'm a final-year student learning about fintech investing. I sized the SME payments market for a class project and found your team's report really useful. Could I share my 3-line takeaway and ask for your view? Happy to keep it to 10 minutes."

Case Study: Verifying a Startup

The situation. You read in an article: "Razorpay is a $7.5 billion payments leader, growing fast." Before you put this in your investment note, you check every claim:

Claim	How to check it	What you find
Valued at $7.5 billion	Compare Crunchbase with two or more news reports on the December 2021 round.	It matches. Series F of $375 million, co-led by TCV, Lone Pine and Alkeon.
Revenue of ₹3,783 cr in FY25, up 65%	Compare the company statement with Entrackr and Inc42, then look for the official filing.	News reports agree. Filings were still pending when reported, so mark it "partly verified".
Growing fast, so doing well	Read the loss breakdown, not just the headline.	There was a ₹1,209 cr loss, mainly one-off tax and ESOP costs from moving the company back to India.
Handles $180 billion in payments a year	Search for a second, independent source and ask the company for the data.	It is company-stated, so write "unaudited" in your notes.

What a careful analyst learns here. Headlines are a starting point. The loss looks scary until you read why it happened, and the volume number sounds great until you realise only the company has reported it.

A case where verification failed. Theranos was a US health-tech startup that raised hundreds of millions of dollars on big claims about its blood-testing machines. Many investors never had the technology tested by independent experts. It later collapsed and its founder was convicted of fraud. The lesson is simple: check what a founder says with documents and people from outside the company.

Assignment

Pick a real startup that is **not** Razorpay and submit a short document or spreadsheet (3 or 4 pages, with sources listed at the end) with three parts:

- **Part A: Market sizing.** TAM, SAM and SOM with your assumptions written down and a sanity check against the startup's real revenue or users. (From Day 4.)
- **Part B: Due diligence.** The filled checklist plus at least five KPIs, each checked against two sources. Clearly mark anything you could not find.
- **Part C: Mock deal pipeline.** Use Crunchbase and LinkedIn to find 8 startups in one sector. Score them with the scorecard, give each a stage and a next action, and end with your top pick and the reason why.

Quick Recap

- Due diligence means checking a startup's claims before investing, across team, market, product, finances, legal papers, customers and competition.
- Never trust a KPI until you have seen it in two places, and asking for missing numbers is part of the job.
- LinkedIn, Crunchbase, PitchBook, Tracxn and company filings are your main sourcing tools.
- A scorecard-based pipeline keeps leads organised, and networking brings deals early and helps you verify what founders say.

Sources: Razorpay FY25 results as reported by Entrackr, Inc42 and Samco; funding rounds from FinTech Global and Moneycontrol. The pipeline stages and scores are made up for practice.

Appendix: Verified Reference Links & Data Sources

This supplementary page provides direct URL links and official web platforms for all data sources, databases, financial news publications, and regulatory bodies referenced across Modules 4 and 5 (Startup Ecosystem, Market Sizing, Unit Economics, and Due Diligence).

Source / Platform	Description / Coverage	Official URL Link
Crunchbase	Funding rounds, investors, similar companies, and sector lists	https://www.crunchbase.com/
PitchBook	Detailed private market deal, valuation, and VC data	https://www.pitchbook.com/
Tracxn	Sector maps, competitor lists, and Indian startup intelligence	https://tracxn.com/
Entrackr	News on funding rounds and yearly financial results of Indian startups	https://entrackr.com/
Inc42	Deep coverage of the Indian startup ecosystem and market trends	https://inc42.com/
FinTech Global	Global FinTech information services and B2B media intelligence	https://fintech.global/
Moneycontrol	Financial news, market analysis, and economic reports	https://www.moneycontrol.com/
LinkedIn	Founder backgrounds, team size tracking, and professional networking	https://www.linkedin.com/
Reserve Bank of India (RBI)	Regulatory body for payment aggregators and financial institutions	https://www.rbi.org.in/
NPCI	National Payments Corporation of India (UPI infrastructure)	https://www.npci.org.in/
Startup India / DPIIT	Government initiatives, policy support, and startup recognition	https://www.startupindia.gov.in/